

Is Hyperbole a Scalar Inference?

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Abstract. This article is concerned with the interpretation of hyperbolic statements. In the first part, we contend that Kao, Wu, Bergen, and Goodman’s (2014)—to our knowledge, the first (and only) attempt to derive hyperbolic interpretations using formal methods—is unsuccessful. In the second part, we put forth a conjecture—roughly, a sentence S can be interpreted hyperbolically as meaning S^- , where S^- is weaker than S , only if S^- is a scalar alternative of S . We substantiate this conjecture by a series of novel empirical observations.

1 Introduction

A hyperbole is an exaggerated statement that is not intended to be taken literally and is typically used to convey a positive or negative feeling, or express a positive or negative evaluation. For example, one may utter ‘Chomsky wrote 10,000,000 books’, despite being common ground that Chomsky didn’t write 10,000,000 books, to communicate that he wrote a lot of books and that one is positively impressed by this. To the best of our knowledge, Kao, Wu, Bergen, and Goodman (2014) stands alone as the only attempt to derive hyperbolic interpretations using formal methods.¹ In this article, we argue that this attempt is unsuccessful. In addition, we advance a conjecture that, if proven true, would establish a fundamental connection between the interpretation of hyperbolic statements and the computation of scalar implicatures.

2 Kao, Wu, Bergen, and Goodman (2014)

Kao, Wu, Bergen, and Goodman (2014) put forth a model of hyperbole interpretation within the Rational Speech Act (RSA) framework (Frank and Goodman 2012). For ease of reference, we will refer to this model as ‘HRSA-14’ (for ‘Hyperbole RSA, 2014’). This model assumes

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¹ Kao et al. (2014) is also presented as an account of ‘pragmatic halo’ effects in natural language (Lasnik 1999). In this article, our sole focus is on Kao et al. (2014) as an account of hyperbole.

that utterance interpretation operates along two dimensions (formalised as QUDs): the state-of-the-world dimension (e.g. *What’s the weather like?*, *How much money does Chris owe?*, *Is John coming to the party tomorrow?*) and the speaker-affect dimension (*Is the speaker affected?*). In addition, it assumes that the speaker chooses her utterances to maximise the probability of accomplishing her goals, which can be communicating information along the state-of-the-world dimension, along the speaker-affect dimension, or along both.

Bergen (2016: 145) provides an example—using the input sentence ‘Bob owes me \$1,000,000’—that illustrates how HRSA-14 works. In Bergen’s example, there are 3 possible money states (\$10, \$100, and \$1,000,000), corresponding to the amount of money that Bob owes the speaker, and 2 possible affect states (T for negative affect and $\perp$ for absence of affect). This yields 6 worlds, each a money–affect pair ($M = x$, $A = y$), where x is the amount Bob owes the speaker and y is the speaker’s affect. The speaker is assumed to know both how much Bob owes her and her own affective state. The listener’s prior (also the common prior²) is assumed to satisfy two constraints: (i) $P(A = T \mid M = \$10) < P(A = T \mid M = \$100) < P(A = T \mid M = \$1,000,000)$ (i.e. larger debts make it more likely the speaker is unhappy); and (ii) $P(M = \$1,000,000) < P(M = \$100) \leq P(M = \$10)$ (i.e. a million-dollar debt is less likely than smaller debts). The prior values Bergen uses in his simulation are given below.

x	$P(M = x)$	$P(A = T \mid M = x)$
\$10	0.495	0.1
\$100	0.495	0.33
\$1,000,000	0.01	0.9

Table 1. The column labelled $P(M = x)$ provides the marginal probability of each possible money state, while the column labelled $P(A = T \mid M = x)$ provides the conditional probability of the speaker having affect given each possible money state.

Bergen (2016) assumes that the speaker is in world (100, T), wants to communicate affect information (i.e. address the QUD *Is the speaker affected?*) and, to do so, can utter either ‘(Bob owes me) \$10’, ‘(Bob owes me) \$100’, or ‘(Bob owes me) \$1,000,000’ (the literal meaning of these utterances corresponds to the set of worlds compatible with states \$10, \$100, and \$1,000,000, respectively). According to Table 1, ‘\$1,000,000’ best serves the speaker’s goal, and thus she utters that sentence. The listener, for her part, has to work out whether the speaker’s goal is to communicate affect or, alternatively, state-of-the-world information—e.g.

² Like most game-theoretic models of communication, Kao et al. (2014) exploits the *common prior assumption*—for an overview, see Morris (1995).

whether the speaker aims to address *Is the speaker affected?* or *How much money does Bob owe the speaker?* Because it is highly unlikely that Bob owes the speaker \$1,000,000, the listener favours the former hypothesis. She therefore takes the utterance of ‘\$1,000,000’ not as a literal claim about the amount owed, but as a way of signalling that the speaker is affected (in this case, negatively).

This example illustrates the general idea behind HRSA-14: to communicate affect, a speaker can utter a declarative sentence S —even if she knows S is false—provided that (i) S is very unlikely to be true given what is known, and (ii) S does a good job at communicating information along the speaker-affect dimension. Given these conditions, upon hearing S , the listener will discard the hypothesis that the speaker wants to communicate that S is the case and will conclude instead that the speaker’s goal is to communicate her affect.

The listener, importantly, will typically also extract state-of-the-world information from hyperbolic statements; as Bergen (2016: 149) points out, this is because the listener often has prior knowledge that the state of the world and the speaker’s affect are linked. In the present example, if the speaker has affect T (negative affect), it is much more likely that she is in a \$100-world than in a \$10-world—that is, in a world where Bob owes her a significant amount of money rather than very little.

2.1 A critical assessment of HRSA-14

HRSA-14 represents a significant achievement in formal pragmatics: it provides an explicit, empirically testable treatment of a phenomenon that had long resisted formal analysis. However, upon examining the model’s predictions, at least three problems become apparent. First, while the model delivers correct results under certain choices of priors, it fails under other equally reasonable choices (§2.1.1). Second, the model is structurally unable to tell apart plain falsehoods from genuine instances of hyperbole (§2.1.2). Third, the model fails to predict the attested hyperbolic interpretation of sentences such as ‘She solved an unsolvable problem’, i.e. sentences whose literal meaning is necessarily false (§2.1.3). In what follows, we discuss these three problems in turn.

2.1.1 The water is boiling!

In Bob’s case (discussed in the previous section), HRSA-14’s reasoning can be summarised as follows:

(i) The speaker utters *Bob owes me \$1,000,000*. (ii) ‘This can’t be true!’, the listener says to herself. ‘The speaker is surely trying to communicate something else other than state-of-the-world information.’ (iii) ‘If the speaker was in a world in which *Bob owes me \$1,000,000* was true’, the listener reasons, ‘she’d be, almost certainly, unhappy.’ (iv) ‘Voilà’, the listener concludes, ‘the speaker is trying to tell me she’s very unhappy (i.e. the speaker is trying to communicate her affect).’ (v) ‘And, surely, if the speaker is very unhappy, then Bob must owe her a significant amount.’

It is worth noting that this reasoning succeeds in highly idealised configurations in which the speaker’s affective state can be explained only by reference to a single state-of-the-world dimension—in Bergen’s example, the money-owed dimension; in Kao et al.’s (2014) simulations, the price dimension—and, moreover, in which there is only *one* likely reason for the speaker to be affected: the world being somewhat high (or somewhat low) on that dimension. In Bergen’s example, the only likely reason for affect is Bob owing the speaker a significant amount of money; in Kao et al.’s simulations, the only likely reason is the price of the relevant item being too high.

In what follows, we consider a slightly more complex case: one in which, as in Kao et al. (2014), the speaker’s affective state is causally linked to a single state-of-the-world dimension but, unlike in Kao et al. (2014), there are *two* likely reasons that can account for that affect—namely, the world being somewhat high as well as the world being somewhat low on that dimension. As we will see, this creates a problem for the model.

Take the utterance ‘The water (in the bathtub) is boiling’. In most situations, it will not convey that the bathtub water is literally boiling, but will be interpreted hyperbolically as meaning, roughly, ‘The water (in the bathtub) is very hot’. If one tries to run the reasoning in (i)–(v) with this utterance, however, a problem arises at step (v):

(i) The speaker utters *The water in the bathtub is boiling*. (ii) ‘This can’t be true!’, the listener says to herself. ‘The speaker is surely trying to communicate something else other than state-of-the-world information.’ (iii) ‘If the speaker was in a world in which *The water in the bathtub is boiling* was true’, the listener reasons, ‘she’d be, almost certainly, unhappy.’ (iv) ‘Voilà’, the listener concludes, ‘the speaker is trying to tell me she’s very unhappy (i.e. the speaker is trying to communicate her affect).’ (v) ‘And, surely, if the speaker is very unhappy, then the bathtub water must be **either too cold or too hot** to take a bath.’

The problem is that at step (v) the listener can conclude only that the water is either too cold or too hot (since the speaker’s unhappiness could be due to either extreme). But this conclusion is too weak: in practice, an utterance like ‘The water (in the bathtub) is boiling’ yields a much

stronger inference—namely, that the water is *very* hot (too hot to take a bath). The simulations reported in the Appendix (§6) provide support for the assessment just given.

2.1.2 King won the Nobel

HRSA-14 is global in nature: as far as the uttered sentence is concerned, the model sees only its literal meaning—that is, the proposition it expresses. On this account, therefore, the following contrast cannot be explained:

- (1) a. This exercise is impossible to solve. (H ✓)
 b. This exercise doesn't have a solution. (H X)

(1)a and (1)b are (arguably) truth-conditionally equivalent: both entail that the exercise has no solution.³ Despite this, (1)a, but not (1)b, supports hyperbole.⁴

Furthermore, given the reasoning that HRSA-14 implements, it is not clear how to account for the fact that (2)a, but not (2)b, works as an exaggeration ((2)b is just a plain lie).

- (2) a. Stephen King won every (literary) prize. (H ✓)
 b. Stephen King won the Nobel Prize (in literature). (H X)

Indeed, according to HRSA-14, it is possible to interpret (2)b hyperbolically—like (2)a, it is extremely unlikely to be true and evokes high-affect worlds. In fact, in the Appendix (§6), we show that, under a natural choice of priors, HRSA-14 predicts (2)a and (2)b to have the same hyperbolic interpretation—namely, that Stephen King won many prizes. This is clearly not the right prediction.

Thus, insofar as one accepts that ‘Stephen King won the Nobel Prize’ does not work as an exaggeration (unlike ‘Stephen King won every prize’), then HRSA-14’s predictions are

³ Unless ‘impossible’ is understood relative to an agent ‘impossible [for us/for a human being/for an AI/etc.]’.

⁴ If the reported judgment isn’t immediately clear, consider (i) and (ii). In (i), B’s utterance ‘it’s impossible to solve’ can clearly be understood hyperbolically (as meaning ‘it’s very difficult to solve’). In (ii), by contrast, B appears to contradict herself, which indicates that her utterance ‘it doesn’t have a solution’ cannot be interpreted as meaning ‘it’s very difficult to solve’.

- (i) A: Prof. Smith is mad. The exercise he gave us is really difficult. I’ve been trying to solve it for 5 hours to no avail.
 B: I know. Stop trying. It’s impossible to solve. The good news is: my dad, who is a mathematician, knows how to do it. Shall I tell you?
- (ii) A: Prof. Smith is mad. The exercise he gave us is really difficult. I’ve been trying to solve it for 5 hours to no avail.
 B: # I know. Stop trying. It doesn’t have a solution. The good news is: my dad, who is a mathematician, knows how to do it. Shall I tell you?

obviously problematic. What the model appears to lack is a principle that distinguishes plain lies (e.g. ‘Stephen King won the Nobel Prize’, ‘Erdős proved the Riemann hypothesis’) from sentences that give rise to a hyperbolic (non-literal) interpretation (e.g. ‘Stephen King won every prize’, ‘Erdős proved thousands of theorems’).

2.1.3 He solved an unsolvable problem!

Probabilistic conditioning is at the heart of HRSA-14: the listener learns about the affective state of the speaker by conditioning the common prior on the proposition expressed by a factual utterance such as ‘Bob owes me \$1,000,000’. In the light of this, the following two utterances, both of which can be understood hyperbolically, constitute a problem for this approach:

- (3) a. He broke an unbreakable chair. (*≈ He broke a chair that is very hard to break.*)
 b. He solved an unsolvable problem. (*≈ He solved a problem that is very hard to solve.*)

Indeed, (3)a-b, on any reasonable semantics, are necessary falsehoods (in probability terms, $P(\llbracket(3)a\rrbracket) = P(\llbracket(3)b\rrbracket) = 0$). Thus HRSA-14 cannot derive the attested interpretations for these cases, since conditioning on a zero-probability proposition is not possible; such an operation is undefined.

One reaction is to see such failures as reflecting a general limitation of possible-worlds semantics, rather than a problem intrinsic to HRSA-14: since, on this semantics, necessarily false sentences all collapse onto the same meaning, any theory that adopts it will inevitably be too coarse-grained when distinctions among necessarily false sentences matter.⁵ This interpretation is defensible. But there is another: the real issue lies not (primarily) with possible-worlds semantics, but with the model’s architecture. Indeed, in HRSA-14, deriving a non-literal interpretation requires first updating the common ground with the utterance’s literal meaning; as a result, in cases in which the literal meaning has probability zero, the derivation cannot get off the ground. It is important to stress, however, that this is a design feature of HRSA-14, not an unavoidable commitment of a theory of hyperbole or, more generally, of non-literal interpretation. In what follows, we explore an alternative idea, couched in a possible-worlds semantics, that does not commit one to HRSA-14’s update-first-literal-meaning architecture.

⁵ Thanks to Leon Bergen for pressing on this point.

Given the problems noted, we do not think that HRSA-14 can be regarded as a successful theory of hyperbole. At best, it is incomplete; at worst, it is incorrect. In the next section, we will advance a theoretical conjecture, one that links the phenomenon of hyperbole with that of scalar implicatures.

3 Is hyperbole a scalar inference?

Hyperbole can be viewed, at least in part, as a weakening problem. Assume, for example, that there is a jar containing chocolate and vanilla cookies. How is it that ‘Julia ate all of the cookies in the jar’, if it is common ground that Julia could not have possibly eaten all of them, can be hyperbolically interpreted as meaning *Julia ate most of the cookies in the jar*, but not as, say, *Julia ate all of the chocolate cookies in the jar*? Both interpretations are weaker than the original sentence, yet only the former is attested. Tellingly, scalar implicatures (SIs) give rise to a structurally analogous problem (only that, in this case, it is a *strengthening problem*), e.g. how is that ‘Julia ate some of the cookies in the jar’ can be interpreted as meaning *Julia ate some of the cookies in the jar, but it is not the case that she ate all of the cookies in the jar*, but not as *Julia ate some of the cookies in the jar, but it is not the case that she ate all of the chocolate cookies in the jar* (both interpretations are stronger than the original sentence, yet only the former is attested).

Most attempts at solving the problem that SIs give rise to⁶ have been guided by the following conjecture:

- (4) An utterance of a sentence S can be understood as conveying $S \wedge \neg S^+$, where S^+ is stronger than S , only if S^+ (or a sentence contextually equivalent to S^+)⁷ is a scalar alternative of S .

⁶ A particular way of framing this problem came to be known as the ‘symmetry problem’ (Kroch 1972; Fox 2007; Katzir 2007, among others).

⁷ The parenthetical is needed to protect the conjecture from counterexamples of the following kind. Under standard assumptions, (ii), but not (iii), is a scalar alternative of (i):

- (i) Some of the diaries of Dante Alighieri were hidden in an old brewery.
- (ii) All of the diaries of Dante Alighieri were hidden in an old brewery.
- (iii) All of the diaries of the author of the Divine Comedy were hidden in an old brewery.

If an utterance of (i) can be understood as conveying ‘(i) and not (ii)’, then, given that (ii) and (iii) are contextually equivalent, it can also be understood as conveying ‘(i) and not (iii)’. Without the parenthetical, the fact that (i) can be understood as conveying ‘(i) and not (iii)’ would constitute a counterexample, since (iii) is not an alternative of (i). As the conjecture is formulated here, however, this is not a counterexample: although not itself an alternative of (i), (iii) is contextually equivalent to (ii), which, under standard assumptions, does qualify as an alternative of (i).

From this perspective, the reason why ‘Julia ate some of the cookies in the jar’ cannot be interpreted as meaning ‘Julia ate some of the cookies in the jar, but it is not the case that she ate all of the chocolate cookies in the jar’ lies in the fact that ‘she ate all of the chocolate cookies in the jar’—unlike ‘she ate all of the cookies in the jar’—does not qualify as a scalar alternative of the uttered sentence. Here the term ‘scalar alternative’ is used in a theoretically neutral sense: an alternative that (neo-)Gricean theories of pragmatic reasoning must, at some level, assume to be available in order to account for a given scalar implicature.

Here we would like to propose an analogous conjecture concerning the interpretation of hyperbolic statements:

- (5) An utterance of a sentence S can be understood hyperbolically as conveying S^- , where S^- is weaker than S , only if S^- (or a sentence contextually equivalent to S^-) is a scalar alternative of S .

(5) enables us to answer the question raised at the outset of this section (how is it that ‘Julia ate all of the cookies in the jar’, if it is common ground that Julia could not have possibly eaten all of them, can be hyperbolically interpreted as meaning *Julia ate most of the cookies in the jar*, but not as, say, *Julia ate all of the chocolate cookies in the jar?*) in a straightforward manner: ‘Julia ate most of the cookies in the jar’ is a scalar alternative of ‘Julia ate all of the cookies in the jar’; ‘Julia ate all of the chocolate cookies in the jar’, by contrast, is not.

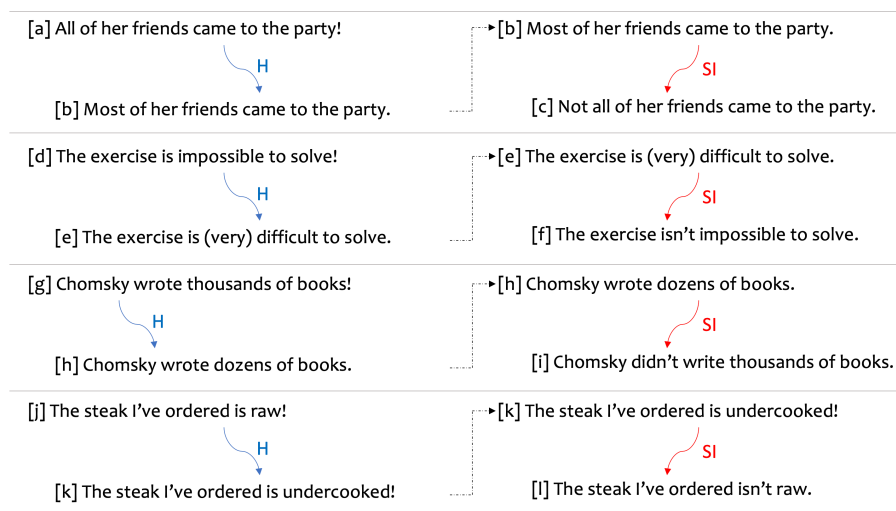


Fig. 1. The ‘H’ arrow associates a (false) statement S with α , a scalar alternative of S , whose meaning can be viewed as a possible hyperbolic interpretation of S . The ‘SI’ arrow associates α with $\neg S$.

In addition, (5) makes two interesting predictions. According to (5), if S can be interpreted hyperbolically as meaning S^- , then S must have a scalar alternative α that is contextually

equivalent to S^- (α may, of course, be S^- itself). This leads us to expect the following: (i) it should be possible to paraphrase the hyperbolic interpretation that S gives rise to by using α ; (ii) if S is itself a scalar alternative of α ,⁸ an utterance of α would be expected to have the scalar implicature $\neg S$ (because S is stronger than α). These predictions appear to be borne out (see Fig. 1).⁹

More generally, if (5) holds, it is reasonable to expect parallel behaviour between hyperbole and scalar implication, at least within certain paradigms. In what follows, we show that important parallelisms between these two phenomena can indeed be found.

3.1 Parallelism I (Numerals)

On neo-Gricean approaches to number interpretation, ‘Three boys blah’ is (typically) analysed as having weak truth-conditions (*three or more boys blah*), with the upper-bounded component (*no more than three boys blah*) derived as a scalar implicature (e.g. Horn 1972; Schulz and van Rooij 2006). From this perspective, the difference between (6)a and (6)b is that the former implicates that Chomsky did not write more than 10,000,000 books, whereas the latter entails it. (In what follows, we assume that this analysis is correct.)

- (6) a. Chomsky wrote 10,000,000 books.
 $\rightsquigarrow_{SI}$ Chomsky didn’t write more than 10,000,000 books.
 b. Chomsky wrote exactly 10,000,000 books.
 $\rightsquigarrow_{SI}$ Chomsky didn’t write more than 10,000,000 books.
 $\Rightarrow$ Chomsky didn’t write more than 10,000,000 books.

Now consider (7) (where n is a high yet plausible number in the context of book writing, e.g. 40): whereas (7)a supports hyperbole, (7)b does not.¹⁰

⁸ In a framework such as Horn’s (1972), this is always the case, as the scalar-alternative relation is symmetric.

⁹ Of course, to arrive at a definite conclusion, a much larger dataset would need to be tested. Fig. 1 is only meant as a preliminary assessment of the proposed conjecture.

¹⁰ If the judgment is not immediately clear, consider the contrast between (iii) and (iv):

- (iii) A: How many books did Chomsky write?
 B: I don’t know... The man wrote 10,000,000 books.
 (iv) A: How many books did Chomsky write?
 B: #I don’t know... The man wrote exactly 10,000,000 books.

(iii)-B is clearly interpreted hyperbolically, as meaning ‘I don’t know... The man wrote a huge number of books.’ (iv)-B, by contrast, does not lend itself to this interpretation and is understood literally instead—indeed, unlike (iii)-B, (iv)-B is perceived as incoherent.

- (7) a. Chomsky wrote 10,000,000 books.
 $\leadsto_H$ Chomsky wrote n books or more.
 b. Chomsky wrote exactly 10,000,000 books.
 $\leadsto_H$ Chomsky wrote n books or more.

(6) and (7) thus exhibit parallel patterns. In its non-hyperbolic use, ‘Chomsky wrote 10,000,000 books’ implicates that Chomsky did not write more than 10,000,000 books; ‘Chomsky wrote exactly 10,000,000 books’, by contrast, does not—indeed, it entails (and thus does not implicate) that Chomsky did not write more than 10,000,000 books. Similarly, ‘Chomsky wrote 10,000,000 books’ can be interpreted hyperbolically as meaning ‘Chomsky wrote n books [or more]’, where n is high but plausible; ‘Chomsky wrote exactly 10,000,000 books’, by contrast, cannot. If one accepts the proposed conjecture and further assumes that the scalar alternatives of ‘Chomsky wrote (exactly) 10,000,000 books’ can be derived only by replacing ‘10,000,000’ with another bare numeral, this parallelism is unsurprising: ‘Chomsky wrote exactly 10,000,000 books’, unlike ‘Chomsky wrote 10,000,000 books’, has neither stronger nor weaker alternatives. Hence it is neither expected to be strengthened via a scalar implicature nor to receive a hyperbolic interpretation.

‘Chomsky wrote n books [or more]’, it should be noted, is not the most natural paraphrase of (7)a’s hyperbolic interpretation—compared to, say, ‘Chomsky wrote a lot of books’, which feels more appropriate. Under the proposed conjecture, this is unexpected: ‘Chomsky wrote n books [or more]’ is a scalar alternative of (7)a, whereas ‘Chomsky wrote a lot of books’ is not (at least under standard assumptions), nor is it contextually equivalent to a scalar alternative of (7)a.¹¹

Does this observation threaten the conjecture? We do not think so, at least not fundamentally. If one turns the conjecture into a procedural account of how the listener recovers the message communicated by a hyperbolic statement, one gets something like this: interpreting a sentence S hyperbolically amounts to identifying its content not with $\llbracket S \rrbracket$ but with $\llbracket S^- \rrbracket$, where S^- is a weaker scalar alternative of S (and plausibly true given common knowledge). But upon hearing ‘Chomsky wrote 10,000,000 books’, interpreters face many potential landing sites for the hyperbole inference, each corresponding to a scalar alternative of the uttered sentence (e.g. ‘Chomsky wrote 40 books [or more]’, ‘41 books [or more]’, ‘42 books [or more]’, etc.). Which one should be selected? In a more realistic model of communication, the hyperbolic

¹¹ Indeed, unless one picks a highly contrived context, ‘a lot of books’, unlike ‘ n or more’, would be expected to have a vague lower-bound.

interpretation of a sentence like (7)a would be identified not with a single scalar-alternative meaning (a proposition), but with a distribution over possible scalar-alternative meanings. From this perspective, it makes sense that, in describing the observed interpretation of (7)a, one reaches for a vague sentence (e.g. ‘Chomsky wrote a lot of books’, ‘Chomsky wrote a vast number of books’): such paraphrases capture what no individual scalar alternative does—namely, that after interpreting (7)a hyperbolically, one cannot be sure what the lowest number of books Chomsky definitely wrote is.¹²

3.2 Parallelism II (Indirect Implicatures)

As discussed in §2.1.2, (1)a—repeated as (8)a below—can be used/interpreted hyperbolically; by contrast, (1)b—repeated as (8)b below—cannot.

- (8) a. This exercise is impossible to solve.
 $\neg_H$ This exercise is (very) difficult to solve.
 b. This exercise doesn’t have a solution.
 $\neg_H$ This exercise is (very) difficult to solve.

Now, consider (9)a–b, the negations of (8)a–b, respectively:

- (9) a. This exercise isn’t impossible to solve.
 $\neg_{SI}$ This exercise is (very) difficult to solve.
 b. This exercise has a solution.
 $\neg_{SI}$ This exercise is (very) difficult to solve.

(9)a carries the (indirect) implicature that the exercise is difficult (or very difficult) to solve; (9)b, by contrast, does not. The observed parallelism, if viewed through the lens of the proposed conjecture, is not at all surprising: (9)a, and not (9)b, is interpreted as meaning ‘The exercise isn’t impossible but it’s (very) difficult to solve’ because (9)a, unlike (9)b, has the alternative ‘The exercise isn’t (very) difficult to solve’; likewise, (8)a, and not (8)b, is interpreted hyperbolically as meaning ‘The exercise is (very) difficult to solve’ because (8)a, unlike (8)b, has the alternative ‘The exercise is (very) difficult to solve’.

It should be noted that the most accurate paraphrase of (8)a’s hyperbolic interpretation is ‘The exercise is *very* difficult to solve’ (or ‘The exercise is *extremely* difficult to solve’), and not

¹² For the connection between vague sentences and the communication of probabilistic information, see Égré et al. (2023).

the plain ‘The exercise is difficult to solve’. Under standard assumptions, however, alternatives cannot be syntactically more complex than the uttered sentence.¹³ One option, therefore, is to maintain that the relevant alternative is ‘The exercise is difficult to solve’ (whatever this means in the context in which (8)a is uttered¹⁴), and explain the ‘very’-component of the inference in terms of affect-based probabilistic enrichment—roughly, ‘The exercise is impossible to solve’ is a strong cue that the speaker has negative affect; on the assumption that affect is due to the exercise’s difficulty, it is more likely than not that the difficulty is quite high. Another option is to consider recent proposals for which an increase in syntactic complexity is less of a barrier to alternativehood, thus allowing a sentence like ‘The exercise is *very* difficult to solve’ to count, in principle, as an alternative of (8)a (e.g. Schwarz and Wagner 2024; Haslinger and Schmitt 2025). Irrespective of the route one takes, the crucial point for present purposes is to expose the following parallelism: both (8)a and (9)a lead the listener to infer that the exercise is quite difficult (via weakening in the former case and via strengthening in the latter), whereas the truth-conditionally equivalent (8)b and (9)b do not.

It is also worth noting that a recent line of work has suggested that the notion of complexity that matters for alternatives is semantic rather than syntactic, tying it in particular to monotonicity (see, especially, Buccola et al. 2022 and Enguehard 2025). This view is *prima facie* plausible: non-monotonic expressions are, in certain aspects, more complex than monotonic ones. Experimentally, non-monotonic quantifiers have been shown to be harder to learn than monotonic quantifiers (e.g. Steinert-Threlkeld and Szymanik 2019; Chemla et al. 2019); and theoretically, measures of semantic complexity have been proposed under which non-monotonic quantifiers come out more complex than monotonic ones (e.g. Carcassi et al. 2019; van de Pol et al. 2023). On this view, ‘very difficult’ can straightforwardly be treated as an alternative of (9)a: although ‘difficult’ and ‘very difficult’ differ in overt (syntactic)

¹³ This desideratum is built into Katzir’s (2007) account, and it is a consequence of Horn’s (1972) proposal—or at least of the way his proposal is usually understood. Indeed, if Horn scales are lexical in nature—and they are standardly taken to be—then alternatives can be generated only by replacing scalar lexical items with other scalar lexical items (e.g. replacing ‘impossible’ with ‘very difficult’ should not be possible in principle).

¹⁴ As is known, relative adjectives like ‘difficult’, ‘tall’ or ‘rich’ express properties only relative to contextually-provided thresholds or comparison classes.

complexity, they pattern alike with respect to monotonicity—both are upward monotonic¹⁵—and hence can be said to be of equal semantic complexity.¹⁶

3.3 Parallelism III (Hurford Disjunctions)

According to Hurford's (1974) well-known generalisation, often referred to as Hurford's constraint (HC), a sentence that contains a disjunctive phrase is odd if one of the disjuncts entails the other.¹⁷ This generalisation correctly predicts (10)a–b to be odd; however, as Hurford himself noted, it fails to make sense of the fact that (10)c is oddness-free ('[Mary] solved both problems' entails 'Mary solved the first problem or the second problem').

- (10)a. # John bought a computer or a laptop.
 b. # John either killed or murdered Paul.
 c. Mary solved the first problem or the second problem, or she solved both problems.¹⁸

Chierchia, Fox, and Spector (2008) argued that (10)c is not a counterexample to HC:¹⁹ on their view, (10)c and other apparent violations of this generalisation involve the presence of a implicature-computing operator *exh* within the first disjunct that 'breaks' the entailment relation between the disjuncts—due to *exh*, the first disjunct is strengthened to 'Mary solved the first problem or the second problem but not both', and, as a result, the sentence does not violate HC.

Another standard case of apparent HC violation, discussed in Gazdar (1979) as well as in Chierchia, Fox, and Spector (2008), can be exemplified with a sentence such as 'Jane read either some of the books or all of the books.' Under the assumption that 'all' presupposes that

¹⁵ If an object *x* falls in the extension of '(very) difficult', then any object *y* such that *y* is equally or more difficult than *x*, will also fall in the extension of '(very) difficult'. Because of this, '(very) difficult' can be said to denote a monotonically increasing function.

¹⁶ Note that the account just outlined immediately makes sense of the contrast between (vii) and (viii):

- (vii) Is this exercise difficult, or is it very difficult?
 (It reads as 'Is this exercise difficult but not very difficult, or is it very difficult?' This is consistent with the view that 'very difficult' is a scalar alternative of 'difficult'.)
- (viii) # Is this exercise difficult, or is it difficult but not very difficult?
 (It doesn't read as 'Is this exercise very difficult, or is it difficult but not very difficult?' This is the reading that one would expect if 'difficult but not very difficult' was a scalar alternative of 'difficult'.)

¹⁷ 'entails' has to be understood under its generalised version, so that it can apply to pairs of non-propositional constituents.

¹⁸ The example is from Chierchia, Fox, and Spector (2008).

¹⁹ Hurford (1974) did not regard (10)c as a counterexample either; rather, he took it to be evidence that in English 'or' is ambiguous between an exclusive and an inclusive reading, a view that gives rise to a number of well-known problems—see Gazdar (1979) and Chierchia, Fox, and Spector (2008) for discussion.

its restrictor is non-empty, ‘all of the books’ entails ‘some of the books’; despite this, Hurford oddness is not observed in such a case. On Chierchia, Fox, and Spector’s (2008) view, this is because *exh* strengthens the first disjunct to ‘some but not all of the books’, thereby breaking the entailment relation between the disjuncts.

Now, one may ask: why isn’t (10)a and (10)b also rescued by *exh*? That is, why is it that the first disjuncts of (10)a and (10)b are not strengthened to ‘computer but not laptop’ and ‘killed but not murdered’, respectively? This question, to our knowledge, has not yet received a conclusive answer. What one may call the ‘received’ view has it that the *is-an-alternative-of* relation is mediated by the notion of *lexical scale* (hence the name *scalar alternatives*; Horn 1972, Gazdar 1979): whereas certain words (e.g. ‘or’/‘and’, ‘some’/‘all’) form a *lexical scale* (and hence are alternatives of one another), others (e.g. ‘computer’/‘laptop’, ‘kill’/‘murder’) do not. For this to constitute an answer to the question raised, however, one would need a general and reliable method capable of determining which pairs of words count, and which do not count, as lexical scales, something we currently lack (recall Gazdar’s (1979:58) famous statement: “scales are, in some sense, ‘given to us’”).²⁰

Here we will not be concerned with this question; we will simply take the contrast between (10)a–b and (10)c to be a manifestation of the following generalisation: if β entails α and ‘ α or β ’ is not odd (i.e. the first disjunct is strengthened to ‘ α and not β ’), then β —or some expression

²⁰ Katzir (2007), notably, takes a different approach and puts forth an explicit account of alternative generation based on considerations of complexity. Though this work makes sense of a great deal of data that was left unexplained in Horn’s (1972) framework, it still fails to offer a principled explanation of why a structure such as ‘kill or murder’ is odd, whereas ‘some or all’, for example, is not. For Katzir (2007), ‘some or all’ is good because ‘all’ is an alternative of ‘some’ (they are of the same complexity) while ‘some but not all’ (the so-called *symmetric* alternative) is not (it is not of the same complexity as ‘some’). Due to this, local exhaustification can proceed and HC ends up not being violated. But this analysis, as noted in Feinmann (2025), leads to the prediction that ‘kill or murder’ should be good too: ‘murder’ is an alternative of ‘kill’ (they are of the same complexity) while ‘unintentionally kill’ (the symmetric alternative) is not (it is more complex than ‘kill’). Even if one were to endorse the view that ‘murder’ is represented at LF as ‘intentionally kill’, Katzir’s account would not be out of trouble: for one thing, it is not clear that such an assumption helps—according to Katzir (2007), a complex expression can serve as an alternative to a simpler expression provided the former is a linguistic constituent (and ‘murder’ is a linguistic constituent in ‘kill or murder’; see Katzir’s (2007) solution to Matsumoto’s problem); furthermore, if ‘murder’ were to be represented at LF as ‘intentionally kill’, then ‘John didn’t murder Paul’ would be predicted to implicate that John killed Paul—this implicature, however, does not seem to be there, as the contrast between (v) and (vi) reveals:

- (v) A: John didn’t intentionally kill Paul.
B: Wait... Has John killed Paul?! I thought Jane had killed him.
- (vi) A: John didn’t murder Paul.
B: # Wait... Has John killed Paul?! I thought Jane had killed him.

(v)-A clearly gives rise to the inference that John killed Paul, hence the naturalness of (v)-B; (vi)-A, by contrast, does not seem to give rise to the same inference—if it did, (vi)-B should be as felicitous as (v)-B, and it clearly isn’t.

contextually equivalent to β^{21} —is a scalar alternative of α ; conversely, if β entails α and ‘ α or β ’ is odd, then β is not a scalar alternative of α . Then, if (5), the proposed conjecture, is correct, and the scalar-alternative relation is taken to be symmetric, the following prediction can be made: for any ‘bad’ Hurford disjunction H , it should never be possible to use H ’s strong disjunct to hyperbolically convey that H ’s weak disjunct is the case. This prediction, as illustrated in Fig. 2, is borne out.

Bad Hurford disjunctions	Hyperbole unavailable
[a] # Has John eaten a piece of fruit or an apple ?	[b] John has eaten an apple . $\nwarrow$ H $\searrow$
	[c] John has eaten a piece of fruit .
[d] # John bought a computer or a laptop .	[e] John bought a laptop . $\nwarrow$ H $\searrow$
	[f] John bought a computer .
[g] # John either killed or murdered Paul.	[h] John murdered Paul. $\nwarrow$ H $\searrow$
	[i] John killed Paul.

Fig. 2. Testing the following prediction: for any ‘bad’ Hurford disjunction H , it is never possible to use H ’s strong disjunct to hyperbolically convey that H ’s weak disjunct is the case.

It should be noted that, if one takes ‘good’ Hurford disjunctions as reference point, the implicature data and the hyperbole data, as shown in Fig. 3, are not entirely parallel, even if one focuses on cases in which the strong disjunct can be more or less uncontroversially taken to be a scalar alternative of the weak disjunct (see fn. 21).

The fact that this is so, it should be noted, does not threaten (5). According to the proposed conjecture, a sentence S can be interpreted hyperbolically as meaning S^- *only if* (and not *if and only if*!) S^- (or a sentence contextually equivalent to S^-) is a scalar alternative of S . That is, the proposed conjecture entails that having a weaker scalar alternative is necessary for S to be interpreted hyperbolically; it does not entail, however, that this condition is sufficient, i.e. it does not commit us to the view that, whenever such a weaker alternative exists, a hyperbolic interpretation must be available. The kind of example that would pose a threat to (5) is a ‘bad’ Hurford disjunction whose strong disjunct can be used hyperbolically to convey that its weak disjunct is the case—such cases, as far as we can tell, do not exist (see Fig. 2).

²¹ This provision is needed for much the same reason as in (4). If β entails α and ‘ α or β ’ is oddness-free, two different things may be going on—(i) β is a scalar alternative of α , or (ii) there is a sentence β' , distinct from β but equivalent in meaning, that is a scalar alternative of α . (If the arguments provided in §3.2 are correct, the oddness-free disjunction ‘Either the exercise is (very) difficult to solve or it is impossible to solve’ is a case of (i), whereas the oddness-free disjunction ‘Either the exercise is (very) difficult to solve or it doesn’t have a solution’ is a case of (ii).) Conversely, if β entails α and ‘ α or β ’ is odd, then β can confidently be taken *not* to be a scalar alternative of α .

Good Hurford disjunctions	Hyperbole available
[a] ✓ Is the exercise (very) difficult or impossible?	[b] This exercise is impossible. [c] This exercise is (very) difficult.
[d] ✓ Don't eat beef that is (very) undercook or raw.	[e] The beef is raw. [f] The beef is (very) undercooked.
[g] ✓ John read most or all of the books.	[h] John read all of the books. [i] John read most of the books.
	Hyperbole unavailable
[j] ✓ John read 1 or 2 books.	[k] John read 2 books. [l] John read 1 book.
[m] ✓ John solved either ex. 1 or ex. 2, or ex. 1 and ex. 2.	[n] John solved ex.1 and ex. 2. [o] John solved ex. 1 or ex. 2.

Fig. 3. ‘Good’ Hurford disjunctions whose stronger disjunct gives rise to hyperbole ([a], [d], [g]) and ‘good’ Hurford disjunctions whose stronger disjunct does not give rise to hyperbole ([j], [m]).

3.4 Comparison with Kao et al. (2014)

In §2.1, we criticised Kao et al.’s (2014) framework on the basis of three data points, namely:

- (11) a. The water in the bathtub is boiling.
 $\rightsquigarrow_H$ b. The water in the bathtub is (very) hot (i.e. too hot for a bath).
 $\rightsquigarrow_H$ c. The water in the bathtub is either too hot or too cold (i.e. not at an appropriate temperature for taking a bath).
- (12)a. Stephen King won every (literary) prize.
 $\rightsquigarrow_H$ b. Stephen King won many (literary) prizes.
c. Stephen King won the Nobel Prize (in literature).
 $\rightsquigarrow_H$ d. Stephen King won many (literary) prizes.
- (13) a. He solved an unsolvable problem.
 $\rightsquigarrow_H$ b. He solved a (very) hard/difficult problem.

Let’s start with (11). Given an utterance of (11)a, Kao et al.’s (2014) model predicts an interpretation that can be paraphrased (at least in part) by (11)c, thus failing to derive the intuitively correct interpretation in (11)b. If the conjecture given in (5) is on the right track, (11)a would not be expected to give rise to the interpretation in (11)c: to our knowledge, no approach to alternatives predicts ‘too cold or too hot’ to be a scalar competitor of ‘boiling’. This is desirable, since otherwise an utterance ‘The water is *not* boiling’ would be (incorrectly) predicted to trigger the (indirect) implicature that the water is either too cold or too hot. Furthermore, on the proposed conjecture, there is no obstacle to assigning (11)a the hyperbolic

interpretation in (11)b: (11)b is weaker than (11)a, hence all that needs to be assumed is that (11)b is a scalar alternative of (11)a, something for which there is independent support (e.g. ‘The water is *not* boiling’, at least in many contexts, does trigger the (indirect) implicature that the water is (very) hot).

Let’s turn now to (12). As discussed, and as shown in the Appendix, Kao et al.’s (2014) model predicts that (12)a and (12)c can both be understood as conveying that King won many literary prizes (a bad prediction). This prediction does not follow from the conjecture proposed here: (12)d is not weaker than (12)c—as a matter of fact, these two sentences are logically independent. Thus, on the proposed conjecture, (12)c cannot be hyperbolically interpreted as (12)d. Conversely, the fact that (12)a can be understood as (12)b is compatible with our proposal: (12)b is weaker than (12)a; moreover, ‘every’ and ‘many’ are standardly taken to be scalar competitors (‘not every’ can be used to convey ‘many’, and ‘many’ can be used to convey ‘not every’).²²

Finally, consider (13)a. This example poses a problem for Kao et al.’s (2014) account: although (13)a can be used hyperbolically (to convey roughly (13)b), Kao et al. (2014) cannot generate this interpretation. As discussed in §2.1.3, the issue is that deriving a hyperbolic interpretation in their system requires first updating the common prior with the uttered sentence’s literal meaning. This operation cannot be carried out for (13)a, because $P(\llbracket(13)a\rrbracket) = 0$ ((13)a is a necessary falsehood).

From the perspective of the proposed conjecture, the fact that (13)a is a necessary falsehood is irrelevant: (13)a can be understood as (13)b because (13)b is a scalar alternative of (13)a—indeed, ‘The problem is not unsolvable’ typically carries the (indirect) implicature that the problem is (very) hard. More concretely, consider a simple procedural account compatible with our conjecture on which interpreting (13)a hyperbolically amounts to substituting ‘unsolvable’ with ‘hard’ (or with ‘very hard’, assuming that the addition of an intensifier preserves scalar

²² A reviewer points out that, while they agree that (12)c cannot be understood hyperbolically as conveying that Stephen King won many prizes, it might nonetheless support a different hyperbolic interpretation—namely, that Stephen King won an important prize. We do not think that this interpretation is available. As far as we can tell, it is not possible to use ‘X won the Nobel Prize (in Y)’ to convey the weaker claim that X won some important prize, not necessarily the Nobel Prize. If that were possible, one could say of a highly decorated physicist like Stephen Hawking, who did not win the Nobel Prize, ‘Hawking won the Nobel Prize in Physics’ to mean simply that he won some major prize. By contrast, one can readily say of Hawking ‘He was awarded 1,000,000 prizes’ to convey that he was awarded many prizes—though of course not literally one million. That said, these are just our intuitions, which do not agree with the reviewer’s. If future experimental work shows that the reviewer’s proposed interpretation for (12)c is in fact available, that would pose a problem for our proposal. Indeed, if the reviewer’s proposed interpretation exists, then ‘Stephen King won an important prize’, according to our conjecture, must be a scalar alternative of (12)c. This, however, seems unlikely: assuming that the scalar-alternative relation is symmetric, the oddness of ‘Either Stephen King won an important prize or the Nobel Prize in Literature’ indicates otherwise.

competition) at LF. Such an account derives (13)b without updating the common ground with (13)a.

Of course, the fact that the data in (11)–(13) do not challenge the proposed conjecture does not mean that the conjecture *explains* these data. The conjecture is a generalisation, and generalisations are not theories. By contrast, Kao et al.’s (2014) is a theory—one that aims to explain hyperbole. Thus, we do not want the conjecture to be viewed as an alternative to Kao et al.’s (2014) account. It is not. RSA theorists could instead try to refine Kao et al.’s (2014) model so as to capture the conjecture/generalisation (and thereby avoid at least some of the problematic predictions discussed in §2.1). This would presumably require revising the model’s architecture so that hyperbole interpretation is constrained by the (un)availability of scalar alternatives. How to implement this technically—assuming it is feasible—we leave for future work.

3.5 Caveats and limitations

The first thing to note is this: the proposed conjecture contributes to solving what we have termed *hyperbole’s weakening problem*, but only partially. Consider, for example:

(14) Julia ate all of the cookies in the jar, and she watched her favourite movie.

(14) can be interpreted hyperbolically as meaning (i) ‘Julia ate most of the cookies in the jar (, and she watched her favourite movie)’, but not as meaning (ii) ‘Julia ate all of the cookies in the jar or she watched her favourite movie’. Note that both (i) and (ii) are scalar alternatives of (14) and are weaker than (14); the question raised in §3 therefore re-emerges: why can (14) be interpreted hyperbolically as (i) but not as (ii)? The proposed conjecture is silent on this.

It seems clear that not all alternatives can serve as landing sites for the hyperbole inference (e.g. ‘*p* and *q*’ cannot be interpreted hyperbolically as ‘*p* or *q*’). In other words, being a weaker alternative is not sufficient to qualify as a possible interpretation of a hyperbolic statement. As already noted in relation to the hybrid results in Fig. 3, this does not threaten the proposed conjecture: (5) is compatible with the observation that not all scalar alternatives are hyperbole-friendly. It remains for future work to determine what further constraint(s)—in addition to (5)—are required to provide a complete solution to hyperbole’s weakening problem.

It is also worth noting that a theory of hyperbole should do more than just solve hyperbole’s weakening problem: it should also explain the source of hyperbole’s expressive or evaluative component. Kao et al. (2014) set out to meet both of these challenges at once, and they deserve credit for that; ultimately, that is the kind of account that is needed. Our aim here, however, has been more modest: to advance a generalisation that may help address the weakening problem.

Finally, we would like to consider a class of potential counterexamples to the proposed conjecture. Take, for example:

(15) John is a tower.

It seems uncontroversial that (15) can convey that John is much taller than average. It is not clear, however, that (15) has a scalar alternative that could be recruited to generate this interpretation. Suppose it does not. Should we then treat (15), or related examples such as ‘John is a giraffe’ or ‘John is a pyramid’, as counterexamples to the proposed conjecture?

We are not persuaded that we should. There are at least two coherent ways of analysing (15) that are (or can be made) compatible with (5). One is that ‘John is a tower’ is a metaphor, and thus an instance of a different phenomenon. Another is that it is both a metaphor and a hyperbole:²³ ‘John is a tower’, a category mistake if John is taken to be a person, could first receive a metaphorical interpretation—suppose, for simplicity, that the metaphor mechanism replaces ‘a tower’ with ‘(at least) 30m tall’. Then ‘John is (at least) 30m tall’, which is almost certainly false, can be interpreted hyperbolically as meaning, for example, ‘John is (at least) 2m tall’, which (under standard assumptions) is a scalar alternative of ‘John is (at least) 30m tall’. (See §3.1 for thoughts on why ‘John is (very) tall’ would typically be preferred over ‘John is (at least) 2m tall’ as a paraphrase of (15)’s hyperbolic interpretation.)

Of course, these two analyses make sense provided that the classical picture—on which hyperbole and metaphor are distinct phenomena that may or may not co-occur—is on the right track. If that picture turns out to be wrong, then (15) may indeed pose a problem for the proposed conjecture.

²³ The view that hyperbole and metaphor can co-occur is already found in Aristotle. ‘Well-liked hyperboles are also metaphors; for example, of a man with a black eye, — “You would have thought him a basket of mulberries”; for his face is somewhat purple, but there is much exaggeration.’ (Aristotle [ca. 335 BCE] 2007: 225).

4 Nouwen (2024)

In recent work, Nouwen (2024) proposed a generalisation pertaining to the conditions under which a hyperbolic interpretation can be expected to arise. Since Nouwen’s generalisation and the conjecture advanced here were developed more or less in parallel,²⁴ we limit ourselves to discussing the main structural differences between them, leaving an in-depth comparison for future work.

At the heart of Nouwen’s generalisation is the claim that hyperbole is mediated by the presence of a contextual scale, which he formalises as a contextually salient *ordering* over QUD cells. It is in this sense that, for Nouwen, hyperbole is a *scalar* phenomenon. The formal details of his proposal are as follows. Context is represented as a triple (Q, n, h) , where Q is a QUD, a partition of the set of worlds compatible with the interlocutors’ beliefs, which, in the cases of interest, is taken to be order-inducing, i.e. its cells are assumed to be ordered by a relation $\preceq$ (Nouwen does not spell out a general procedure for deriving $\preceq$, which is taken to be contextually given). The parameter n is the ‘norm/expectation’ QUD cell, and h is the QUD cell containing the actual world. For a sentence S , Nouwen defines $\tau_Q(S)$, viz. the set of Q ’s cells compatible with (the literal meaning of) S . With this machinery in place, he advances the following generalisation:²⁵

(16) **Hyperbole** (Nouwen 2024: 710): An utterance of S in (Q, n, h) counts as hyperbole iff

- (i) $n \prec h$,
- (ii) for every c in $\tau_Q(S)$, $h \prec c$ and
- (iii) the scalar distance between h and c is large.²⁶

(In words: an utterance of S counts as hyperbole iff, relative to the contextual ordering, the actual-world cell h lies beyond the norm cell n , and the literal content of S places matters still further in that direction, by a large margin.)

To see how this definition works, consider Nouwen’s example (minimally adapted). John invites 60 people and expects about 30 of them to come, but in fact 58 show up. Thus, in such a case, $n = \{30\}$ and $h = \{58\}$. The QUD is taken to be the degree question *How many people attended the party?*; accordingly, Q is the partition $\{\{0\}, \{1\}, \{2\}, \dots\}$, where each cell

²⁴ The first version of the present article, available on LingBuzz, dates from March 2023; the article by Nouwen discussed here, available on the same archive, dates from January 2024.

²⁵ Nouwen also offers a generalisation for meiosis, another figure of speech. Here we focus only on his account of hyperbole.

²⁶ Nouwen does not tell us how large it has to be in order to count as hyperbole; that is an aspect of the formal framework that is left intuitive.

corresponds to an exact number of attendees. The ordering is assumed to be the natural one, i.e. higher attendance ranks higher. This gives us the following context: $(Q, n, h) = (Q, \{30\}, \{58\})$, with $\{30\} < \{58\}$.

John, to express surprise at the high attendance, utters ‘There were 100 people in the living room!’. Let’s compute τ_Q (‘There were 100 people...’): if ‘100’ is ascribed an exact reading, the literal content of ‘There were 100 people ...’ is compatible only with the cell $\{100\}$; thus, τ_Q (‘There were 100 people ...’) = $\{\{100\}\}$.²⁷ In the stated context, ‘There were 100 people ...’ should count as an instance of hyperbole—indeed, all three conditions in (16) are satisfied:

- (i) $n < h$: $\{30\} < \{58\}$ ✓
- (ii) for every c in $\tau_Q(S)$, $h < c$: since $\tau_Q(S) = \{\{100\}\}$, we have $\{58\} < \{100\}$ ✓
- (iii) the scalar distance between h and c is (intuitively) large ✓

The first thing to note is that the sense in which, for Nouwen, hyperbole is ‘scalar’ is different from the one we have been advocating for. Nouwen’s ‘scale’ is essentially a structured QUD-space: a QUD Q partitions a set of worlds into cells corresponding to possible answers, and hyperbole requires that these cells be equipped with an ordering relation, which he assumes to be contextually given. On our view, what is scalar about hyperbole is that the phenomenon is parasitic on scalar alternatives (in the relevant, alternative-theoretic sense). The word ‘scalar’ in ‘scalar alternatives’ is there mostly for historical reasons, i.e. Horn (1972) famously proposed that such alternatives are derivable from lexical entailment scales. In approaches to alternatives that do not make reference to lexical scales, such as Katzir (2007) and Fox and Katzir (2011), sometimes the expression ‘*formal* alternatives’ is preferred. From our perspective, whether alternatives are called ‘scalar’ or ‘formal’ does not matter (nor does it matter which account of alternatives one thinks is on the right track): what matters is the substantive claim that hyperbole, like standard cases of Quantity-based implicatures (*some* → *not all*, *warm* → *not hot*, etc.), is an alternative-sensitive phenomenon—a claim that, once

²⁷ If ascribed an at-least reading, τ_Q (‘There were 100 people ...’) contains the cell $\{100\}$ together with all higher cells; since all of them lie well above $\{58\}$ on the relevant ordering, the predictions are the same. Although, in the present example, it does not matter which reading is assigned to ‘There were 100 people ...’, the fact that the account yields the same predictions for exact and at-least readings of numerals (when the numeral is high enough) is potentially problematic. For example, it is unclear to us how, if one assumes Nouwen’s framework, the difference in interpretation between (7)a and (7)b could be accounted for.

considered alongside standard diagnostics for alternativehood, yields specific predictions (see, for example, Fig. 1 and Fig. 2 in §3).²⁸

This difference in perspective does have empirical consequences; consider, for example, the following exchange:

- (17) A: Were the boys hungry?
B: They ate all the cookies from the jar!

Here (17)B, which functions as an emphatic ‘Yes!’, can naturally be heard hyperbolically (roughly: they ate most of the cookies, though not necessarily all of them), especially in contexts where it is common ground that they could not have eaten every single cookie—for instance, if the jar is transparent and a few cookies can be seen still sitting at the bottom. From Nouwen’s perspective, however, there should not be hyperbole here. Indeed, (17)B induces a 2-cell QUD, i.e. $\{\{hungry\}, \{not\ hungry\}\}$, whereas Nouwen’s generalisation requires a QUD able to distinguish at least *three* relevant positions on the ordering (norm *n*, actuality *h*, and an utterance-compatible cell *c* that lies further in the same direction).

A natural response on Nouwen’s behalf would be that, in such configurations, the operative QUD is not the one induced by the overt question, but an accommodated one—for example, *How many cookies from the jar did the boys eat?* This alternative QUD has the required structure: assuming it is common ground that there were 100 cookies in the jar, each cell can be identified with a cookie-count. It also comes with a natural ordering for ranking the cells—namely, the usual order on the natural numbers. Let’s thus fix that QUD, and consider the following two replies:

- (18) A: How many cookies from the jar did the boys eat?
B₁: They ate all the cookies from the jar!
B₂: The number of cookies they ate from the jar matched the number of cookies that were in the jar.

²⁸ As a matter of fact, according to Nouwen (2024, fn. 3), it is misguided to assume a deep connection between hyperbole and scalar implicatures (in his own words: ‘I think it is wrong to assume such close ties to implicature’.) His point is that ‘I’m dead’ can be understood as meaning ‘I’m very tired’, even though—under standard assumptions—the latter is not a scalar alternative of the former. If one accepts that this is an instance of hyperbole, then it puts pressure on our generalisation, much like example (15) (‘John is a tower’), as well as similar cases such as ‘I cried a river’, ‘This car is a tank’, etc. As discussed in §3.5, it is not obvious that these data should be classified as hyperbole, as opposed to metaphor (or some hybrid of the two). Hence, as things stand, we do not take them to be disqualifying for the proposed conjecture.

In the stipulated context, (18)B₁ and (18)B₂ are equivalent in meaning; as a result, these sentences induce the same $\tau_Q(S)$. Nouwen’s account therefore predicts that they should pattern alike with respect to hyperbole. This prediction, however, is not borne out: whereas (18)B₁ supports hyperbole, (18)B₂ (as far as we can tell) does not.

On the proposed conjecture, the contrast just noted is not a problem in principle. (18)B₁, irrespective of whether one favours Horn (1972)’s approach or the more recent structural approach (Katzir 2007; Fox and Katzir 2011), has the alternative ‘They ate most of the cookies from the jar!’, and hence it can be hyperbolically understood as conveying that. By contrast, no established approach to alternatives that we are aware of predicts (18)B₂ to have ‘They ate most of the cookies from the jar!’ as an alternative. Thus, tentatively, the contrast between (18)B₁ and (18)B₂ can be attributed to the fact that the relevant alternative is available for the former sentence but not for the latter.

It is worth recalling that alternatives, leaving aside the idiosyncrasies of particular implementations, are standardly characterised as alternatives *of a sentence*, not of a proposition, i.e. it is possible for two sentences to have the same meaning yet not the same alternatives. Indeed, since the 1970s, alternatives have been taken to be sensitive to aspects of linguistic representation, whether lexical (Horn 1972) or syntactic (Katzir 2007)—i.e. to how information is encoded in the sentence. Since the conjecture put forward here falls in this tradition, contrasts such as the one between (18)B₁ and (18)B₂ do not undermine it; if anything, they may be seen as lending it support.

5 Final thoughts

Is hyperbole a scalar inference? The answer to this question depends on one’s working definition of ‘scalar inference’. What seems likely is that scalar implicatures and hyperbolic statements share a common element—namely, reliance on *alternatives*.

In this paper, we have tried to avoid as much as possible the issue of what it is that makes a sentence an alternative of another sentence—suffice it is to say that this is a hard problem for which a fully general solution does not yet exist; see Breheny et al. (2018) and Feinmann (2025) for discussion and open challenges. This lack of a general solution, however, does not prevent us from agreeing on whether a given sentence should count (or should not count) as a scalar alternative of another—for example, we agree that ‘All *blah*’ should count as a scalar

alternative of ‘Most *blah*’ because ‘Most *blah*’ has the scalar implicature ‘Not all *blah*’—despite the fact that we might be uncertain about how this alternative ought to be derived. Moving forwards, the research question that we find most exciting is the following: assuming that the proposed conjecture is correct, why is it that not all scalar alternatives can serve as landing site for the hyperbole inference—for example, why is it that ‘*p* or *q*’ has the implicature ‘not(*p* and *q*)’, but ‘*p* and *q*’ cannot be interpreted hyperbolically as meaning ‘*p* or *q*’? Answering this question in a principled way may contribute not only to better understanding how hyperbole works, but also to shedding light on the elusive nature of alternatives.

6 Appendix

Here we present a series of simulations that support our critique of HRSA-14 (see §2). The code for these simulations is available for download here.²⁹

6.1 Problem I: ‘The water (in the bathtub) is boiling!’

HRSA-14 predicts that, under a natural choice of priors, ‘The water (in the bathtub) is boiling’ will be interpreted as meaning that the water is either too cold or too hot to take a bath. In what follows, the priors and other parameters are fixed, and the model’s output is examined.

6.1.1 Model’s parameters

[I] Worlds | 20 in total. Each world consists of a temperature-affect pair ($T = x, A = y$), where x stands for the temperature of the bathtub water, while y stands for the speaker’s affect state. The set of possible temperatures is $\{10n^{\circ}\text{C} : n = 1, 2, \dots, 10\}$, while the set of possible affect states is $\{0, 1\}$ —as in Kao et al. (2014), ‘0’ means without affect, ‘1’ means with affect, and affect, when present, is taken to be negative.

²⁹ <https://osf.io/k6r3f>

[III] Priors | The prior probabilities in Table 2 below were used:

x	$P(T = x)$	$P(A = 1 \mid T = x)$
10°C	0.03	0.9
20°C	0.05	0.8
30°C	0.2	0.7
40°C	0.44	0.01
50°C	0.2	0.75
60°C	0.04	0.8
70°C	0.02	0.9
80°C	0.012	0.97
90°C	0.005	0.98
100°C	0.003	0.99

Table 2. The column labelled $P(T = x)$ provides the marginal probability of each possible temperature state, while the column labelled $P(A = 1 \mid T = x)$ provides the conditional probability of the speaker having affect given each possible temperature state.

The priors are constructed to reflect three background expectations:

- 40°C is the *a priori* most likely water temperature (near body temperature), i.e. people taking baths typically aim for a comfortable temperature rather than an uncomfortable one. Thus, the probability decreases as the water temperature deviates from 40°C: water below 40°C is typically too cold, whereas water above 40°C is typically too hot (Table 2, col. 2).
- Since 40°C is the ideal temperature to have a bath, the probability of the speaker being affected is at its lowest at 40°C, and increases as the temperature moves away from this mark (Table 2, col. 3).
- The priors are not perfectly symmetric to reflect basic constraints and preferences: $P(T = 100^\circ\text{C}) < P(T = 10^\circ\text{C})$ since bathwater cannot reach boiling point in conventional households; and $P(A = 1 \mid T = 10^\circ\text{C}) < P(A = 1 \mid T = 100^\circ\text{C})$, reflecting a preference for enduring 10°C rather than 100°C.

Finally, to avoid introducing bias towards either the cold or hot temperature spectrum, the values in Table 2 (col. 2) were chosen so that the likelihood of the water being too cold for a bath (that is, below 40°C) equalled the likelihood of it being too hot (that is, above 40°C). As can be readily verified, $P(T = 10^\circ\text{C} \vee T = 20^\circ\text{C} \vee T = 30^\circ\text{C}) = P(T = 50^\circ\text{C} \vee T = 60^\circ\text{C} \vee \dots \vee T = 100^\circ\text{C})$.

[III] Messages | 5 in total, with the following meanings:³⁰

- $\llbracket \text{too cold} \rrbracket = \{(T = x, A = y) : x < 40^\circ\text{C}\}$
- $\llbracket \text{OK} \rrbracket = \{(T = x, A = y) : x = 40^\circ\text{C}\}$
- $\llbracket \text{too hot} \rrbracket = \{(T = x, A = y) : x > 40^\circ\text{C}\}$
- $\llbracket \text{boiling} \rrbracket = \{(T = x, A = y) : x = 100^\circ\text{C}\}$
- $\llbracket \text{not OK} \rrbracket = \{(T = x, A = y) : x \neq 40^\circ\text{C}\}$

[IV] QUDs | In HRSA-14, as discussed, utterance interpretation operates along two dimensions (formalised as QUDs)—namely, the state-of-the-world dimension and the speaker-affect dimension.

Like Kao et al. (2014), we used a fine-grained QUD to represent the state-of-the-world dimension—namely, *What’s the temperature of the bathtub water?* Each cell of this QUD corresponds to a distinct temperature state. Unlike Kao et al. (2014), we also tested the model under two coarse-grained state-of-the-world QUDs—namely, *Is the bathtub water too hot, OK, or too cold?* and *Is the bathtub water too hot?* The former induces a three-cell partition: $\{T > 40^\circ\text{C}\}$, $\{T = 40^\circ\text{C}\}$, and $\{T < 40^\circ\text{C}\}$. The latter induces a two-cell partition: $\{T > 40^\circ\text{C}\}$ and $\{T \leq 40^\circ\text{C}\}$. We did this to maximise the model’s chances for success—*a priori* it was conceivable that the model may fail when tested with the fine-grained QUD but succeed with one of the coarse-grained QUDs.

To represent the speaker-affect dimension, we used the same speaker-affect QUD that was used in Kao et al. (2014)—namely, a two-cell partition such that one cell contains all the worlds in which the speaker has (negative) affect and the other all the worlds in which the speaker has no affect.

Kao et al. (2014) included a third (‘hybrid’) QUD—the conjunction of the state-of-the-world QUD and the speaker-affect QUD—to allow for the possibility that the speaker may aim to communicate along both dimensions (see p. 12006, Materials and Methods). We followed Kao et al. (2014) in this respect, but in addition to three-QUD simulations (including the hybrid/conjunctive QUD), we also ran two-QUD simulations (including only the state-of-the-world and speaker-affect QUDs). The rationale, again, was to maximise the model’s chances

³⁰ Each of these messages were assumed to have cost 0. This is in line with Kao et al. (2014: 12007): ‘[Our] analysis of hyperbole doesn’t involve utterance cost’.

for success: *a priori*, it was conceivable that the model might perform poorly with three QUDs yet do well with only two.

Table 3 shows the total number of simulations run (each row is a simulation, hence 6 simulations in total). Simulations whose ID ends in ‘QQ’ only have two QUDs—the speaker-affect QUD and one of the state-of-the-world QUDs; by contrast, simulations whose ID ends in ‘QQQ’ have three QUDs—namely, the speaker-affect QUD, one of the state-of-the-world QUDs, and the conjunction of the previous two QUDs. Thus, the only difference between n QQ and n QQQ simulations is that the latter have an extra QUD, i.e. the conjunction of the two QUDs associated with n QQ.

Simulation ID	Speaker-Affect QUD	State-of-the-World QUD	Affect & State QUD
1QQ	Is the speaker affected?	What’s the temperature of the bathtub water?	×
1QQQ	Is the speaker affected?	What’s the temperature of the bathtub water?	✓
2QQ	Is the speaker affected?	Is the bathtub water too cold, OK, or too hot?	×
2QQQ	Is the speaker affected?	Is the bathtub water too cold, OK, or too hot?	✓
3QQ	Is the speaker affected?	Is the bathtub water too hot?	×
3QQQ	Is the speaker affected?	Is the bathtub water too hot?	✓

Table 3. Full set of simulations ran. The QQ simulations use two QUDs, whereas the QQQ simulations use three (they incorporate an additional ‘hybrid’ QUD that is the conjunction of the speaker-affect QUD and state-of-the-world QUD).

In each simulation, as in Kao et al. (2014), the prior probability over QUDs was assumed to be flat (in QQ-simulations each QUD had prior probability of 1/2, whereas in QQQ-simulations, each QUD had a prior probability of 1/3).

6.1.2 Model’s predictions

HRSA-14 is formulated within the RSA framework, where speaker and listener recursively reason about each other’s intentions to derive pragmatically enriched interpretations (Frank and Goodman 2012). In principle, this recursion can extend to arbitrary depth. At recursive depth 0, the model delivers literal interpretations, which we are not interested in. Non-literal interpretations (hyperbolic interpretations in this case) begin at recursive depth 1. We will be illustrating HRSA-14’s predictions with level-5 interpretations, but we could have done so with

level-1, level-4 or level-6 interpretations as well. To the extent that we were able to verify, all the interpretations obtained at recursive depths greater than 0 are similar enough to support our conclusions.

For a given message or utterance u and recursive depth n , HRSA-14 generates an interpretation of u , which corresponds to the belief state of the n -level listener (formally, a probability distribution over worlds). Fig. 4 (left frame) shows the interpretations that simulations 1QQ and 1QQQ assign to the utterance ‘The water (in the bathtub) is boiling’ at recursive depth 5. As can be seen, these interpretations are nearly identical: after hearing ‘The water (in the bathtub) is boiling’, the listener comes to believe that the water must be either too hot (i.e. above 40°C) or too cold (i.e. below 40°C), and that the speaker is affected (note that the worlds that end in ‘1’ concentrate all the probability mass). Though the affect side of the inference is plausible, the state-of-the-world side is not: upon hearing ‘The water (in the bathtub) is boiling’, the listener should conclude that the water is too hot to take a bath, and not that the water may be too hot to take a bath but may also be too cold.

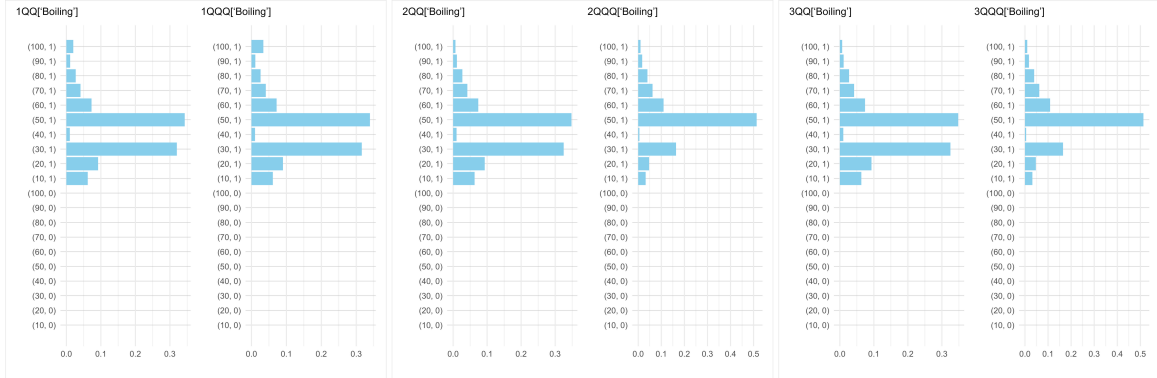


Fig 4. Interpretations for the utterance ‘The water (in the bathtub) is boiling’ according to 1QQ and 1QQQ (left), 2QQ and 2QQQ (centre), and 3QQ and 3QQQ (right). Recursive depth 5, $\lambda = 20$. The vertical axis lists worlds, and the horizontal axis indicates the probability mass assigned to each world.

The same pattern emerges in the other simulations (Fig. 4, centre and right frames). The only novelty is that, with coarse-grained state-of-the-world QUDs, the QQQ (three-QUD) variants perform somewhat better than the QQ (two-QUD) variants. Compare 2QQ and 2QQQ, for example. In 2QQ, given ‘The water (in the bathtub) is boiling’, the posterior that the water is too cold (below 40°C) is 0.48 and that it is too hot (above 40°C) is 0.51. In 2QQQ, the posterior shifts toward the hot (i.e. correct) side of the spectrum: too cold is 0.24 and too hot is 0.75. Despite this improvement, both 2QQQ and 3QQQ still make the wrong prediction: according to these simulations, after hearing ‘The water (in the bathtub) is boiling’, the listener comes to

believe that the water, although more likely than not to be on the hot side of the spectrum, could also be too cold to take a bath.

Another way to assess HRSA-14’s predictions is to compare how the model interprets ‘The water (in the bathtub) is boiling’ vis-à-vis ‘The water (in the bathtub) is too hot’. Intuitively, one would like these two utterances to express the same (or very similar) state-of-the-world (temperature) information: after all, in contexts in which ‘The water (in the bathtub) is boiling’ can be used hyperbolically, ‘The water (in the bathtub) is too hot’ could have been used instead. This isn’t the result that HRSA-14 delivers, however. In Fig. 5 below, we plot the posterior distribution over temperature states (each temperature state is a set consisting of two worlds³¹) induced by ‘The water (in the bathtub) is boiling’ alongside the one induced by ‘The water (in the bathtub is) too hot’.

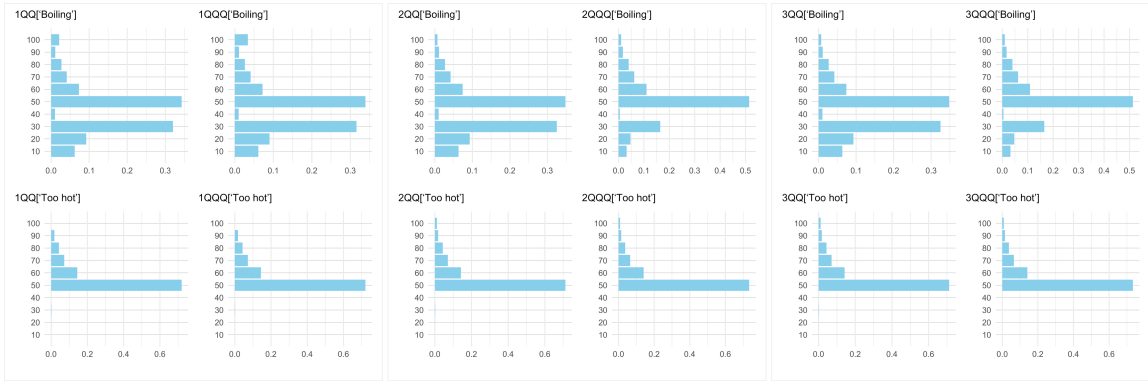


Fig 5. Posterior distributions over temperature states for ‘The water (in the bathtub) is boiling’ (first row) and for ‘The water (in the bathtub) is too hot’ (second row), as delivered by our six simulations (see Table 2). Recursive depth 5, $\lambda = 20$. The vertical axis lists temperature states, and the horizontal axis indicates the probability mass assigned to each state.

‘The water (in the bathtub is) too hot’ is interpreted as it should across all simulations—as shown in Fig. 5, virtually all, if not all, the probability mass lies in the too-hot-for-a-bath range (above 40°C). The water (in the bathtub) is boiling’, by contrast, is not interpreted correctly: in all the simulations it induces a distribution according to which it is entirely possible for the water to be below 40°C.

Furthermore, the interpretation that the model assigns to ‘The water (in the bathtub) is boiling’ is problematic for at least two further reasons. As shown in Fig. 5, the listener (after interpreting ‘The water (in the bathtub) is boiling’) is predicted to come to believe that it is far more likely

³¹ For example, the 50°C temperature state corresponds to the set $\{(T = 50, A = 0), (T = 50, A = 1)\}$.

for the water to be in the 10-40°C range than at 60°C. This prediction does not agree with intuitive judgment: if the water is (hyperbolically) boiling, then it is too hot to take a bath; and that means that it is not in the 10-40°C range but above 40°C. Similarly, the model predicts that, after interpreting ‘The water (in the bathtub) is boiling’, the listener will come to believe that the water is a lot more likely to be at 30°C than to be at 40°C. Again, this makes no intuitive sense: if the water is (hyperbolically) boiling, then it is too hot to take a bath; and, if that is the case, it is absurd to conclude that the water is more likely to be at 30°C (which is 20°C away from the beginning of the too-hot region) than at 40°C (which is only 10°C away).

These results indicate that HRSA-14 is not, as it stands, a successful account of hyperbole interpretation. The model can yield intuitively correct inferences in highly idealised settings where worlds vary along two dimensions and these covary monotonically—e.g. as the state-of-the-world value increases, the probability that the speaker is affected also increases (or at least does not decrease). But when that relation is more complex—as in the bathwater case—its predictions diverge from intuitive judgment.

If the diagnosis just given is correct, the problem exposed in Figs. 4–5 should disappear as soon as the mapping from temperature states to the probability of speaker affect is revised accordingly. To test this, the third column of Table 2 was replaced with 0.001, 0.001, 0.001, 0.001, 0.75, 0.80, 0.85, 0.90, 0.95, 0.99, thereby ensuring that affect varies monotonically with temperature. This adjustment did indeed fix the problem.³² In Fig. 6, we plot the posterior distributions over worlds for 1QQ and 1QQQ that result from running the model with the new ‘friendly’ priors, alongside the corresponding posteriors obtained with the original priors (already plotted in Fig. 4).³³ As can be seen, the posteriors obtained with the friendly priors now concentrate on temperatures above 40°C, in sharp contrast to the posteriors obtained with the original priors.

³² Such priors are compatible, for example, with the following scenario: Imagine John wants to wash some items in the bathtub using special gloves. These gloves are made of a material that should not be exposed to temperatures of 50°C or higher. As the temperature rises above 50°C, the likelihood of the gloves melting and causing harm to John increases.

³³ Virtually identical results were obtained for 2QQ/QQQ and 3QQ/QQQ.

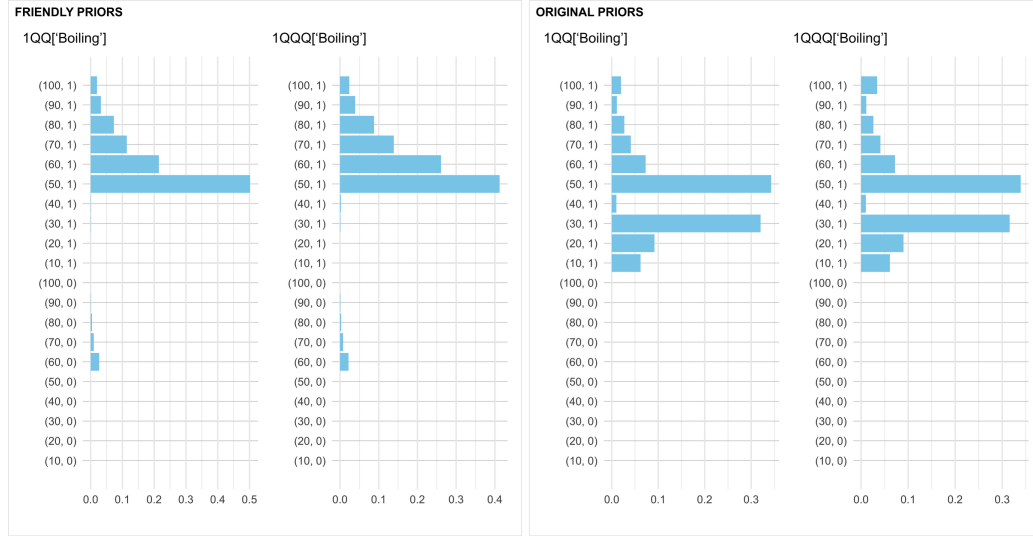


Fig. 6. Interpretations for ‘The water (in the bathtub) is boiling’ according to 1QQ and 1QQQ with ‘friendly’ priors (two plots on the left), and with the original priors (two plots on the right). Recursive depth 5, $\lambda = 20$.

6.2 Problem II: ‘Stephen King won the Nobel Prize!’

To test whether HRSA-14 can capture the contrast between (2)a and (2)b, we ran simulations for both utterances. The input parameters are given below, followed by the model’s predictions.

6.2.1 Model’s parameters

[I] Worlds | We will assume that there are 4 possible prizes: A, B, C (which are prestigious) and the Nobel Prize (notated ‘N’, the most prestigious of all). Each world consists of a pair ($Z = x, A = y$), where x stands for a possible prize combination—e.g. King won A and nothing else (notated ‘Axxx’), King won A and B and nothing else (notated ‘ABxx’)—while y stands for a possible affect state (either ‘1’, which indicates that the speaker has affect (positive affect in this case), or ‘0’ which indicates that the speaker is not affected). Thus, there are a total of 32 worlds (i.e. there are 16 possible prize-combination states and two affect states).

[II] Priors | The priors (see Table 4) were chosen based on the following assumptions (which we take to be entirely reasonable):

- The probability of being in a world in which King won the Nobel Prize is extremely low—as far as anyone knows, he did not.
- The probability of being in a world in which King won no prize is low (it would be highly unusual if an author of such repute had never been awarded a literary prize).

- Given King's fame and reputation, he is more likely to have won two or more prizes as opposed to just one.
- If someone were to learn that King won the Nobel Prize, she would be almost certainly (positively) impressed.
- If someone were to learn that King had won all the prizes in a given set X , plus one additional prize, she would be more likely to be impressed than if she learned that he had won only the prizes in X .

x	$P(Z = x)$	$P(A = 1 \mid Z = x)$
xxxx	0.0104	0.01
Axxx	0.06	0.2
xBxx	0.06	0.2
xxCx	0.06	0.2
xxxN	0.0002	0.99
ABxx	0.202	0.6
AxCx	0.202	0.6
AxxN	0.0002	0.995
xBCx	0.202	0.6
xBxN	0.0002	0.995
xxCN	0.0002	0.995
ABCx	0.202	0.875
ABxN	0.0002	0.9975
AxCN	0.0002	0.9975
xBCN	0.0002	0.9975
ABCN	0.0002	0.99875

Table 4. The prior distribution. The column labelled $P(Z = x)$ provides the marginal probability of each possible prize-combination state, while the column labelled $P(A = 1 \mid Z = x)$ provides the conditional probability of the speaker having affect given each possible prize-combination state.

[III] Messages | 9 in total, with the following meanings:

- $\llbracket \text{every prize} \rrbracket = \{(Z = x, A = y) : x = \text{ABCN}\}$
- $\llbracket \text{the Nobel Prize} \rrbracket = \{(Z = x, A = y) : x \text{ has letter 'N'}\}$
- $\llbracket \text{prize A} \rrbracket = \{(Z = x, A = y) : x \text{ has letter 'A'}\}$
- $\llbracket \text{prize B} \rrbracket = \{(Z = x, A = y) : x \text{ has letter 'B'}\}$
- $\llbracket \text{prize C} \rrbracket = \{(Z = x, A = y) : x \text{ has letter 'C'}\}$
- $\llbracket \text{prize A \& B} \rrbracket = \{(Z = x, A = y) : x \text{ has letters 'A' and 'B'}\}$
- $\llbracket \text{prize B \& C} \rrbracket = \{(Z = x, A = y) : x \text{ has letters 'B' and 'C'}\}$
- $\llbracket \text{prize A \& C} \rrbracket = \{(Z = x, A = y) : x \text{ has letters 'A' and 'C'}\}$
- $\llbracket \text{prize A \& B \& C} \rrbracket = \{(Z = x, A = y) : x \text{ has letters 'A', 'B' and 'C'}\}$

[IV] QUDs | The speaker, as in the previous examples, wants to communicate information along the state-of-the-world dimension and/or along the speaker-affect dimension. The state-of-the-world dimension is identified with the QUD *What prizes has King been awarded?* (a

sixteen-cell partition), and the speaker-affect dimension with the QUD *Is the speaker affected?* (a two-cell partition). We ran two simulations: a QQ simulation, with the state-of-the-world and speaker-affect QUDs, and a QQQ simulation, which additionally includes the conjunction of those QUDs.

6.2.2 Model's predictions

As shown in Fig. 7, both ‘King won every prize’ and ‘King won the Nobel Prize’ were interpreted hyperbolically.

Indeed, in both the QQ and the QQQ simulations, ‘King won every prize’ and ‘King won the Nobel Prize’ induced virtually the same posterior distribution, one which corresponds to a hyperbolic interpretation: (i) the literal meaning of both of these sentences is false (all the worlds in which King won the Nobel Prize, including those in which King won every prize, have probability zero); (ii) the speaker is (positively) affected (the worlds that end in ‘1’ concentrate all the probability mass); (iii) King most certainly won a significant number of prizes (it is more likely than not that King won two or more prizes).

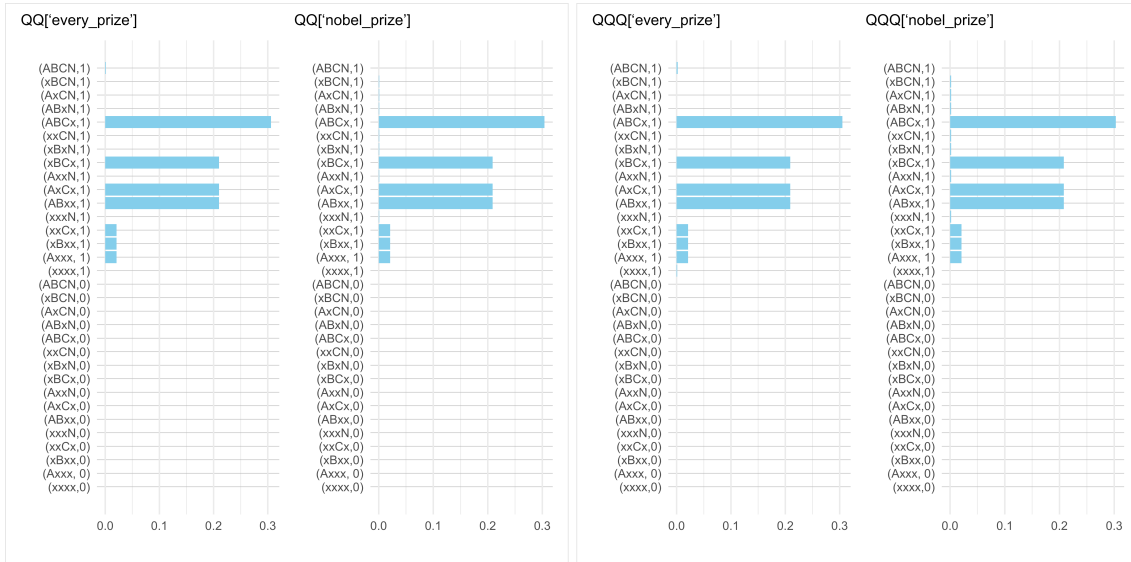


Fig 7. Final interpretations for the utterances ‘King won every prize’ and ‘King won the Nobel Prize’ according to QQ (two plots on left) and QQQ (two plots on right). Recursive depth 5, $\lambda = 20$.

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References

- Aristotle. (ca. 335 BCE) 2007. *On Rhetoric: A Theory of Civic Discourse*. Translated by George A. Kennedy. 2nd ed. OUP.
- Bergen, Leon. 2016. “Joint Inference in Pragmatic Reasoning.” PhD Thesis, Massachusetts Institute of Technology.
- Breheny, Richard, Nathan Klinedinst, Jacopo Romoli, and Yasutada Sudo. 2018. “The Symmetry Problem: Current Theories and Prospects.” *Natural Language Semantics* 26 (2): 85–110. <https://doi.org/10.1007/s11050-017-9141-z>.
- Buccola, Brian, Manuel Križ, and Emmanuel Chemla. 2022. “Conceptual Alternatives.” *Linguistics and Philosophy* 45 (2): 265–91. <https://doi.org/10.1007/s10988-021-09327-w>.
- Carcassi, Fausto, Marieke Schouwstra, and Simon Kirby. 2019. “The Evolution of Adjectival Monotonicity.” *Proceedings of Sinn Und Bedeutung* 23 (1): 219–30. <https://doi.org/10.18148/sub/2019.v23i1.512>.
- Chemla, Emmanuel, Brian Buccola, and Isabelle Dautriche. 2019. “Connecting Content and Logical Words.” *Journal of Semantics* 36 (3): 531–47. <https://doi.org/10.1093/jos/ffz001>.
- Chierchia, Gennaro, Danny Fox, and Benjamin Spector. 2008. “The Grammatical View of Scalar Implicatures and the Relationship between Semantics and Pragmatics.” In *Handbook of Semantics*, edited by Paul Portner, Claudia Maienborn, and Klaus von Heusinger. Mouton de Gruyter.
- Égré, Paul, Benjamin Spector, Adèle Mortier, and Steven Verheyen. 2023. “On the Optimality of Vagueness: ‘Around’, ‘between’ and the Gricean Maxims.” *Linguistics and Philosophy* 46 (5): 1075–130. <https://doi.org/10.1007/s10988-022-09379-6>.
- Enguehard, Émile. 2025. “Towards a Pragmatic Explanation for the Prevalence of Upward-Monotonicity in Natural Language: Some Results on Communicative Stability, the Strongest Answer Condition, and Exhaustification.” *Linguistics and Philosophy* 48 (4): 693–740. <https://doi.org/10.1007/s10988-025-09437-9>.
- Feinmann, Diego. 2025. “On Breaking Symmetry by Complexity.” *Snippets* 47: 8–10.
- Fox, Danny. 2007. “Free Choice and the Theory of Scalar Implicatures.” In *Presupposition and Implicature in Compositional Semantics*, edited by Uli Sauerland and Penka Stateva. Palgrave Studies in Pragmatics, Language and Cognition. Palgrave Macmillan UK. https://doi.org/10.1057/9780230210752_4.
- Fox, Danny, and Roni Katzir. 2011. “On the Characterization of Alternatives.” *Natural Language Semantics* 19 (1): 87–107. <https://doi.org/10.1007/s11050-010-9065-3>.
- Frank, Michael C., and Noah D. Goodman. 2012. “Predicting Pragmatic Reasoning in Language Games.” *Brevia. Science* 336 (6084): 998–998. <https://doi.org/10.1126/science.1218633>.
- Gazdar, Gerald. 1979. *Implicature, Presupposition and Logical Form*. Academic Press.
- Haslinger, Nina, and Viola Schmitt. 2025. “Revisiting the Role of Structural Complexity in Symmetry Breaking.” *Proceedings of Sinn Und Bedeutung* 29, 637–54.
- Horn, Laurence. 1972. “The Semantics of Logical Operators in English.” PhD Thesis, UCLA.
- Hurford, James R. 1974. “Exclusive or Inclusive Disjunction.” *Foundations of Language* 11 (3): 409–11.
- Kao, Justine T., Jean Y. Wu, Leon Bergen, and Noah D. Goodman. 2014. “Nonliteral Understanding of Number Words.” *Proceedings of the National Academy of Sciences* 111 (33): 12002–7. <https://doi.org/10.1073/pnas.1407479111>.
- Katzir, Roni. 2007. “Structurally-Defined Alternatives.” *Linguistics and Philosophy* 30 (6): 669–90. <https://doi.org/10.1007/s10988-008-9029-y>.

- Kroch, Anthony. 1972. "Lexical and Inferred Meanings for Some Time Adverbs." *Quarterly Progress Report of the Research Lab. of Electronics* 104, MIT., 260–67.
- Lasersohn, Peter. 1999. "Pragmatic Halos." *Language* 75 (3): 522–51.
<https://doi.org/10.2307/417059>.
- Morris, Stephen. 1995. "The Common Prior Assumption in Economic Theory." *Economics & Philosophy* 11 (2): 227–53. <https://doi.org/10.1017/S0266267100003382>.
- Nouwen, Rick. 2024. "Meiosis and Hyperbole as Scalar Phenomena." *Proceedings of Sinn Und Bedeutung* 28 (December): 704–18. <https://doi.org/10.18148/sub/2024.v28.1156>.
- Pol, Iris van de, Paul Lodder, Leendert van Maanen, Shane Steinert-Threlkeld, and Jakub Szymanik. 2023. "Quantifiers Satisfying Semantic Universals Have Shorter Minimal Description Length." *Cognition* 232 (March): 105150.
<https://doi.org/10.1016/j.cognition.2022.105150>.
- Schulz, Katrin, and Robert van Rooij. 2006. "Pragmatic Meaning and Non-Monotonic Reasoning: The Case of Exhaustive Interpretation." *Linguistics and Philosophy* 29 (2): 205–50. <https://doi.org/10.1007/s10988-005-3760-4>.
- Schwarz, Bernhard, and Michael Wagner. 2024. "Symmetry Resolution and Blocking." *Proceedings of Sinn Und Bedeutung* 28 (December): 829–47.
<https://doi.org/10.18148/sub/2024.v28.1164>.
- Steinert-Threlkeld, Shane, and Jakub Szymanik. 2019. "Learnability and Semantic Universals." *Semantics and Pragmatics* 12 (November): 4:1–39.
<https://doi.org/10.3765/sp.12.4>.